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COSC 645

2 November 2026

Person-Name Extraction with Support Vector Machines: Applying the CoNLL NER Corpus to the BNC Baby Corpus

Named entity recognition (NER) is the task of finding spans of text that name real-world things and labeling each span with a type such as person, location, or organization (Nadeau and Sekine). This report describes how a support vector machine (SVM) trained on *Conll_NER_corpus.txt* can identify person names in the first 10,000 sentences of the BNC Baby corpus, and reports results from a working implementation.

Named Entities and the Training Data

Conll_NER_corpus.txt is derived from the Annotated Corpus for Named Entity Recognition on Kaggle (Walia) and follows the CoNLL column format (Tjong Kim Sang and De Meulder): one token per line, a tab, and the token's tag, with a blank line between sentences. The file holds 47,959 human-tagged sentences and 1,048,575 tokens. Tags use the BIO scheme (Ramshaw and Marcus): *B-* marks the first word of an entity, *I-* marks a word inside it, and *O* marks a word outside any entity. Person names are tagged *B-per* (16,990 tokens) and *I-per* (17,251 tokens); in this corpus a title such as *President* or *Mr.* is part of the name, so “President Mahmoud Ahmadinejad” is tagged B-per I-per I-per. All other tags (geo, org, gpe, tim, art, eve, nat) are mapped to O, leaving three classes: B-PER, I-PER, and O.

Why an SVM

An SVM finds the hyperplane that separates two classes with the largest margin (Cortes and Vapnik). Linear SVMs handle the very large, sparse feature vectors produced by text well and were a strong NER baseline before neural models (Isozaki and Kazawa). With three tags, one binary SVM is trained per tag (one-versus-rest) and each token receives the tag with the highest score. The implementation uses the LIBLINEAR solver (Fan et al.) through scikit-learn's *LinearSVC* (Pedregosa et al.), and NLTK's *BNCCorpusReader* to read the BNC XML files (Bird et al.).

Procedure

The procedure has two phases, shown in Flowcharts 1 and 2. In training (Flowchart 1), every token becomes a feature dictionary describing the word itself (lowercased form, first and last three letters, capitalization, all-caps, digits, a trailing period as in *Mr.*), its shape (for example, *McCain* becomes *XxXx*), and the same information for the two words on each side, since context such as *said* or a preceding title signals a name. A vectorizer builds a sparse matrix, 10% of sentences are held out to measure precision and recall, and the final model is retrained on all sentences with balanced class weights, because O covers about 97% of tokens.

In extraction (Flowchart 2), the first 10,000 sentences (<*s*> units) of BNC Baby are read in file order, given the same features, and tagged; each B-PER and its following I-PER tokens form a candidate name. A regular expression keeps only spans that look like names (capitalized words, initials, hyphens, apostrophes, and inner particles such as *van* or *de*). Names are written with their sentence numbers, followed by a frequency list.

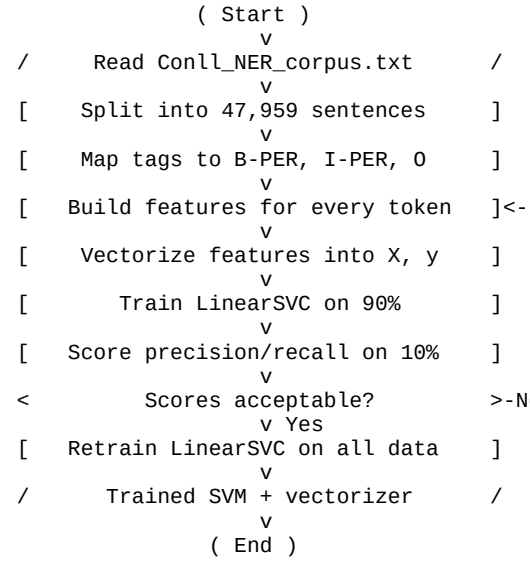
Pseudocode

```
PROCEDURE TrainPersonTagger(conllFile)
    sentences ← ReadSentences(conllFile)          // split at blank lines
    FOR each (word, tag) in sentences:
        tag ← 'B-PER' if tag = 'B-per', 'I-PER' if tag = 'I-per', else 'O'
```

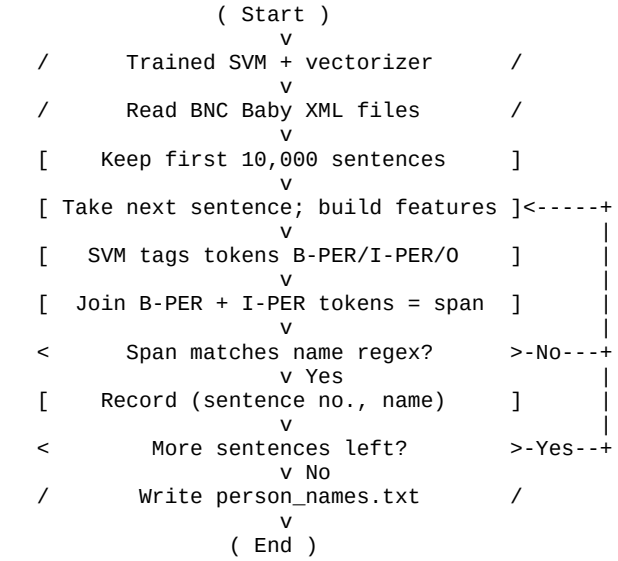
```
    X, y ← [], []
    FOR each sentence s, FOR i ← 0 TO |s|-1:
        X.append(Features(s, i));  y.append(tag of s[i])
    vectorizer ← DictVectorizer().fit(X)
    (Xtr, ytr), (Xte, yte) ← Split(X, y, 90% / 10%)
    svm ← LinearSVC(C = 0.1, class_weight = balanced).fit(Xtr, ytr)
    Report precision, recall, F1 of svm on (Xte, yte)
    RETURN LinearSVC(...).fit(X, y), vectorizer  // retrain on all data

FUNCTION Features(s, i)
    f ← {lower(s[i]), shape(s[i]), prefix3, suffix3, isTitle, isUpper,
         isDigit, endsWithPeriod, isFirstWord}
    FOR k IN {-2, -1, +1, +2}:
        f ← f ∪ {lower(s[i+k]), shape(s[i+k]), isTitle(s[i+k])} or <pad>
    RETURN f

PROCEDURE ExtractPersons(svm, vectorizer, bncDir, N = 10000)
    sents ← first N sentences of BNCCorpusReader(bncDir).sents()
    results ← []
    FOR k, s IN enumerate(sents):
        tags ← svm.predict(vectorizer.transform(Features(s, i) for each i))
        FOR span IN JoinSpans(s, tags):     // B-PER followed by I-PER*
            IF Matches(NAME_REGEX, span): results.append((k, span))
    Write results and name frequencies to person_names.txt
```



Flowchart 1. Training the SVM person-name tagger on the CoNLL corpus.



Flowchart 2. Extracting person names from the first 10,000 BNC Baby sentences.

Results

On the 10% held-out split, token-level scores for B-PER were precision .766, recall .876, and *F*1 .817; for I-PER they were .791, .926, and .853. Applied to the first 10,000 BNC Baby sentences (eight academic texts), the program extracted 1,213 person-name mentions covering 686 distinct names. The most frequent were Levin (20 mentions), Kahlo (16), Debord (16),

Wilson (13), and Lord Scarman (13), alongside false positives such as *Second* (23), *GIS* (12), and *Monte Carlo* (12). As the lab allows, these errors were not corrected.

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